PROJECT REPORT

Submitted to

DEPARTMENT OF COMPUTER SCIENCE

VIVEKANAND EDUCATION SOCIETY'S

COLLEGE OF ARTS, SCIENCE AND COMMERCE

(AUTONOMOUS), SINDHI SOCIETY, CHEMBUR,

MUMBAI 400071.

UNREAL FURNITURE

For Partial Fulfilment for Degree of

Bachelor of Science (Computer Science)

Academic Year (2023-24)

COORDINATOR OF DEPARTMENT

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CERTIFICATE

This is to certify that **Mr. Vikas Ramtirth Jaiswal** of T.Y. B. Sc (Computer Science) affiliated to University of Mumbai has successfully completed a project work entitled.

UNREAL FURNITURE

As partial fulfilment of the requirement for the degree of B.Sc. (Computer Science) for the academic year 2023-2024.

COORDINATOR OF DEPARTMENT

Mr. Kamlakar Bhopatkar

COLLEGE GUIDE

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Ms. Priya Daniel

Date:

Examiner:

College Seal

Acknowledgement:

I have great pleasure in presenting this project entitled "**Unreal Furniture**" and I grab this opportunity to convey my immense regards towards all the people who with their invaluable contributions made this project successful.

It gives me great pleasure in presenting this project report. Its justification will never complete if I don't express my vote of thanks to our **V.E.S. College** and **Principal Dr. Anita Kanwar**

I sincerely thank and express my profound gratitude to our Project Guides **Mr. Kamlakar Bhopatkar and Ms. Priya Daniel** for timely and prestigious guidance required for the project completion at each phase of the project development.

I also owe to my friends who have been a constant source of help to solve the problems that cropped up during the development of the project, positive criticism, suggestions, constant support, encouragement and guidance towards the successful completion of the project.

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UNREAL FURNITURE

Preliminary Investigation

UNREAL FURNITURE

Organizational Overview:

The Furniture industry stands as a sector that helps people by enhancing their experience and interaction with their living areas and workspaces. It provides a wide variety of products for residential spaces like living room, drawing room, bedroom, etc. Beyond our homes, commercial sectors like hotels, offices, educational institutions, etc. also uses wide range of furniture products. In this industry, the main focus is to provide end-users with an ergonomic design, durable, customizable furniture product which is space optimized and is available in suitable price range. Offline retail stores are designed to showcase furniture products allowing consumers to see, touch, and experience the products before making a purchase. These stores are designed to enhance the overall shopping experience, providing a personalized touch that online platforms often struggle to replicate. Although, the rise of e-commerce has revolutionized the way of selling offline furniture products in retail shop. Online Platforms offers end-users the convenience of browsing, selecting, filtering and purchasing furniture from their homes itself. Product images, videos, detailed descriptions which enhances the experience of end-users making a significant improvement in customer engagement. The challenge arises when users struggle to visualize how the furniture will look and fit for their specific spaces. Applications like Ikea Place, Amazon, Wayfair, Houzz, etc. are various online platforms that offers augmented reality (AR) features that allows users to virtually place furniture in their homes or other specific places before making a purchase, enhancing the online shopping experience by providing a realistic preview of how products will look in real-world settings.

Description of Existing System:

- **IKEA Place:**

  IKEA Place is an innovative augmented reality (AR) app by IKEA that allows users to virtually position and visualize furniture in their homes before purchasing. Leveraging AR technology, the app uses a device's camera to create a real-time view of a room, enabling users to select and place virtual IKEA furniture products seamlessly. The app uses lighting, space, etc. to create an accurate image of a product in the user's space. This allows users to see if the product fits their space, style, and purpose. In addition to digitally placing IKEA products in a room, the app allows people to capture the setting in the app and share as an image or video with friends. They can also walk around the furniture to see it as if it were there. IKEA is opening a world of furnishing possibilities to millions of Android users around the world, allowing more people to see how a great piece of furniture might amp up their space. IKEA Place lets users confidently experiment, experience and "place" 3D representations of more than 3,200 IKEA products into their space to ensure that they are just the right size and design. From sofas and lighting to beds and wardrobes. IKEA Place is now available on Android for all AR Core compatible devices.

- **Houzz:**

  Houzz is also an American website, online community, and software for architecture, interior design, decorating, landscape design, and home improvement. The app's commitment to enhancing user experience extends to its 3D visualization capabilities. Users can experiment with different design ideas and product placements, virtually visualizing how these elements would integrate into their own living spaces. The Houzz app serves as a comprehensive marketplace for home products. Users can discover and purchase furniture, decor items, and other essentials directly through the app, facilitating a seamless transition from design concept to implementation. One of Houzz's standout features is its ability to connect users with a diverse array of home improvement professionals. Whether seeking an architect, interior designer, or contractor, users can browse through portfolios and hire professionals directly through the platform. The app's commitment to enhancing user experience extends to its 3D visualization capabilities. Users can experiment with different design ideas and product placements, virtually visualizing how these elements would integrate into their own living spaces.

Limitations of Existing System:

Addressing these limitations is crucial to provide users with a more enhanced and efficient system like "Unreal Furniture"

- Lack of specific product information like availability, reviews and furniture dimensions.
- Limited to specific brand products and thereby providing less options for users.
- Only few products have AR view or a 3-Dimensional view of product.
- Difficulty in tracking progress of product delivery after the purchase of product.
- Limited insights about products and orders for both users and admin.
- The website or app lacks user-friendly design, making it difficult for customers to navigate and shop easily, which can lead to frustration and fewer sales.
- The current system may lack a robust mechanism for collecting user feedback on the overall shopping experience.

Proposed System:

Unreal Furniture is an android application that helps users in efficient furniture shopping through its innovative augmented reality (AR) and 3-Dimensional visualization of products. It provides users with a unique and immersive experience. The platform's AR capabilities allow customers to virtually place furniture products in their own living spaces, aiding in visualizing styles and sizes accurately. Detailed product information, including customer reviews and dimensions, enhances transparency and aids informed decision-making. The commitment to offering AR views for every product ensures a comprehensive and interactive exploration of items in three dimensions, minimizing uncertainties for users. With real-time delivery tracking and efficient customer support, Unreal Furniture prioritizes user satisfaction and re-innovating the way you find your dream furniture.

1. Registration and Sign-in:

- User launches the "Unreal Furniture" app.
- The user is presented with the option to register or sign in if they already have an account.
- If the user is already registered and forgets the password then the user can click on forgot password to reset it.
- The user can also contact the support team in case of any difficulties.
- User must provide valid credentials to explore the application and make a purchase.
- The details of the user will get stored in the firestore.
- After successful registration or sign-in, the user is directed to the home screen.

2. Home Screen Navigation:

- This is the landing screen for the users where the users can view various featured and popular products.
- Users are also provided with various showcase banners displaying new product arrivals and offers going on.
- This screen displays products along with information like prices, ratings, etc.
- User can directly add any product directly to wishlist just by clicking the heart button.

3. Searching and Filtering Products:

- On the search screen, the user will see a search bar and filters to refine their search.
- The user can search directly through the search bar by writing the names of the products.
- The user can also filter products depending upon the prices, ratings and discounts offered.
- The user can apply filters such as category filtering like sofas, beds, tables, etc. and can also view out of stock products.
- The user selects a products to get a detailed view of that product.

4. Viewing Product Details:

- The user is directed to the selected products screen.
- The profile page displays details about the product, including images, description, dimensions, discount, availability and customer reviews and ratings.
- The user can scroll through customer reviews to get an idea of the product’s relevance.
- This screen provides user with various quick action buttons like add to wishlist, cart or the user can directly click on buy now button to make a quick purchase.

- User can also view a 3D model or AR View of the product by clicking on the button that is provided in this screen.

5. **View Products in 3D and AR:**
   - 3-Dimensional Models that allows users to rotate, zoom and view from any angle for a detailed view of furniture product.
   - A robust AR system allowing users to virtually place furniture products in their own living spaces.

6. **Cart and Wishlist:**
   - A user-friendly interface that allows seamless addition of selected furniture products to the shopping cart.
   - Users can also add their favourite products to wishlist to make later purchases.
   - An ability to users to save their current cart sessions, allowing them to revisit and complete purchases at a later time.
   - Enable users to create wishlist based on their likings and allow them to easily transfer items from their wishlist to the cart.

7. **User Profile View:**
   - This screen provides all the information about the current signed in user.
   - The user can update their name, image and shipping addresses without which the checkout is not possible.
   - The current signed-in user can their orders and can also check statuses with detailed information and live tracking of orders.
   - The current user is also provided with a action to cancel the order at any time.
   - App info like policies and version of the application is also provided on this screen.

8. **Checkout Screen:**
   - After confirming the product to purchase, the user can proceed to checkout screen.
   - Here the user can select the shipping address that they have saved in their profile screen.
   - A detailed information of all the selected products, prices, quantities and delivery charges are provided on this screen
   - The user can select a payment method depending upong their preference and complete the checkout.
   - After checkout, the user can easily track the progress of their delivery by accessing the ordered products list and viewing its status.

9. **Notifications:**
   - The app sends realtime-notifications to users regarding various events, new product launches, offers, reviews, delivery tracking, feedback, etc.
   - Provide users with notifications about the status of their order such order placed, order shipped, out for delivery, etc.
   - The app provides notifications to keep users and admins informed about product order updates.

10. **Ratings and Reviews:**
    - After the purchase is completed and the product is delivered, the user receives a notification about the status.
    - The user can provide a review and rating for the product based on their experience.
    - The review is submitted and added to the product's review section for other users to see.

11. **Admin Screen:**
    - Admin have the permissions to add new products by entering product details, images, dimensions, 3D models, etc.
    - Admin can also update the information of any product like its stock, name, description, etc.
    - Admin can also view the orders that are incoming and going on and also have the privilege to change the status of the orders.
    - Admins can also view the insights and analytics to make their business grow.

Advantages of Proposed System:

Some of the Advantages of the Proposed System:

- View product in both 3D and AR view in workspaces.
- Products are available with detailed information like dimensions, description, prices, discounts, etc.
- Convenient customer support and real-time notifications.
- Live tracking of order details with real-time notifications.
- Enhanced wishlist and cart management for later purchases.
- Integration of third-party APIs and libraries, enhances user experience.
- Enhanced user trust and confidence results in retaining customers.

Technologies required for developing the proposed system:

- **Android Studio (Kotlin):** Android Studio is the official Integrated Development Environment (IDE) for Google's Android operating system. Whereas, Kotlin is a High-level programming language. It is one of the official languages of google supported for android application development.

- **Firebase:** Firebase is a comprehensive mobile and web development platform offered by Google. It provides a range of backend services including database management, user authentication, cloud storage, and hosting. Firebase simplifies the development process by offering ready-to-use tools and infrastructure for building high-quality applications.

System Analysis

Use Case

A use case diagram serves as a visual representation of the operations a system undertakes. It comprises key elements such as use cases, actors, and their interconnections. Use cases essentially embody a series of actions that coalesce into a singular unit of work tailored for an actor, who signifies an external user engaging with these operations. In essence, this diagram acts as a pivotal tool for elucidating how a system functions and how external users interact with it, offering a concise yet comprehensive overview of system behaviour.

ELEMENTS OF USE CASE DIAGRAM:

Actor:

Actors in a use case diagram represent entities playing specific roles in a system, typically reflecting real-world user roles. They interact with use cases and should include stakeholders relevant to the system's functionality, with no need to depict entities that don't impact the □odelled functions. Actors are symbolized as stick figures outside the system's boundary.

Use Case:

A use case in a diagram visually represents distinct business functions within a system. To identify use cases, begin by listing discrete business functions as basic scenarios. Each function can be categorized as a potential use case. In a use case diagram, use cases are depicted as ellipses, effectively encapsulating the system's specific functionalities. This approach provides a systematic means of representing the core functions the system executes in response to user interactions or external events.

System Boundary:

The system boundary, depicted as a rectangle in a use case diagram, defines the system's scope by encapsulating all its use cases. It distinguishes the system's internal operations from its interactions with the external environment, aiding in effective system communication and design.

Relationships:

The following relationships can be established among use cases:

Extends:

In a use case diagram, one use case can extend another, indicating that the behaviour of the extension use case can be inserted into the extended use case under specific conditions. This relationship is useful for depicting optional functionalities that come after a particular use case. It's represented by a dashed arrow from the optional use case to the base use case, with the label "«extend»," making it clear that the extended use case can be enhanced or extended with additional features when certain conditions are met.

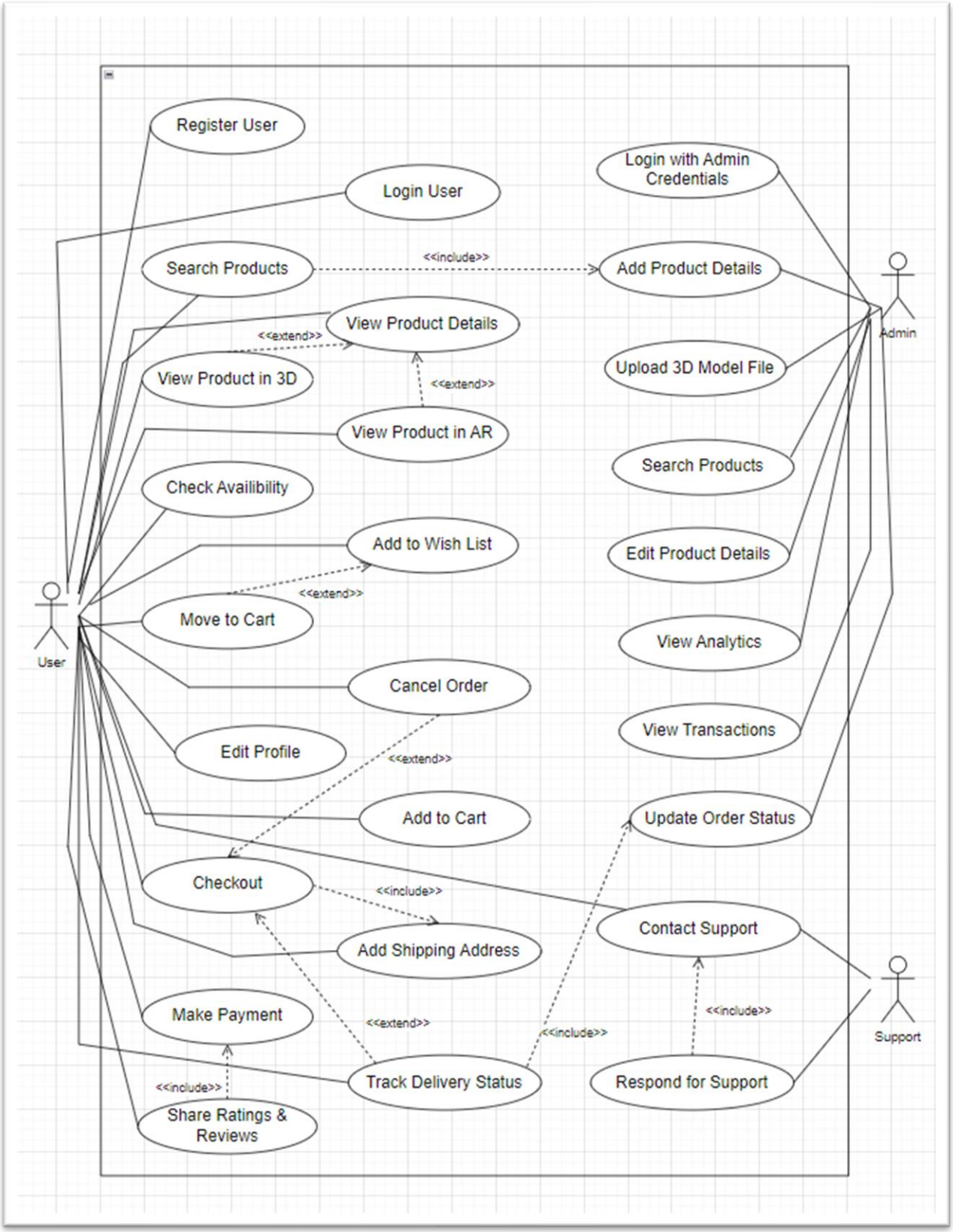
Include:

In a use case diagram, one use case can include another, denoting that the behaviour of the included use case is merged into the behaviour of the including use case. Often, the outcome of the included use case influences the including one. This relationship is represented by a dashed arrow from the base use case to the included use case, labelled "«include»," indicating the integral nature of the included functionality within the encompassing use case.

Basic scenarios for use case

- Registering account
- Logging in
- Add Products
- Update Products
- Search products
- View products
- View products in 3D and AR
- Add to wishlist
- Add to cart
- Add shipping address
- Proceed to checkout
- Track order delivery status
- Receive real-time notifications
- Update order status
- Rate and review products
- Viewing Insights

Use Case Diagram:



Activity Diagram

Activity diagrams, akin to flowcharts, depict the sequence of actions or control flow within a system, commonly used in business process modeling and describing use case steps. They feature sequential and concurrent activities, with a defined start (initial state) and end (final state). These diagrams employ symbols for activities, flows, decisions, guards, merge points, time events, and more, enabling comprehensive visualization of intricate processes and system behaviours.

Initial State or Start Point:
In activity diagrams, a small filled circle followed by an arrow symbolizes the initial action state or the starting point for the diagram's activities.

Activity or Action State:
An action state, representing uninterrupted object actions, is portrayed as a rounded rectangle, facilitating visual clarity in diagrams.

Action Flow:
Action flows, known as edges or paths, delineate transitions between action states in diagrams. Typically depicted as arrowed lines, they clarify the sequential progression of activities.

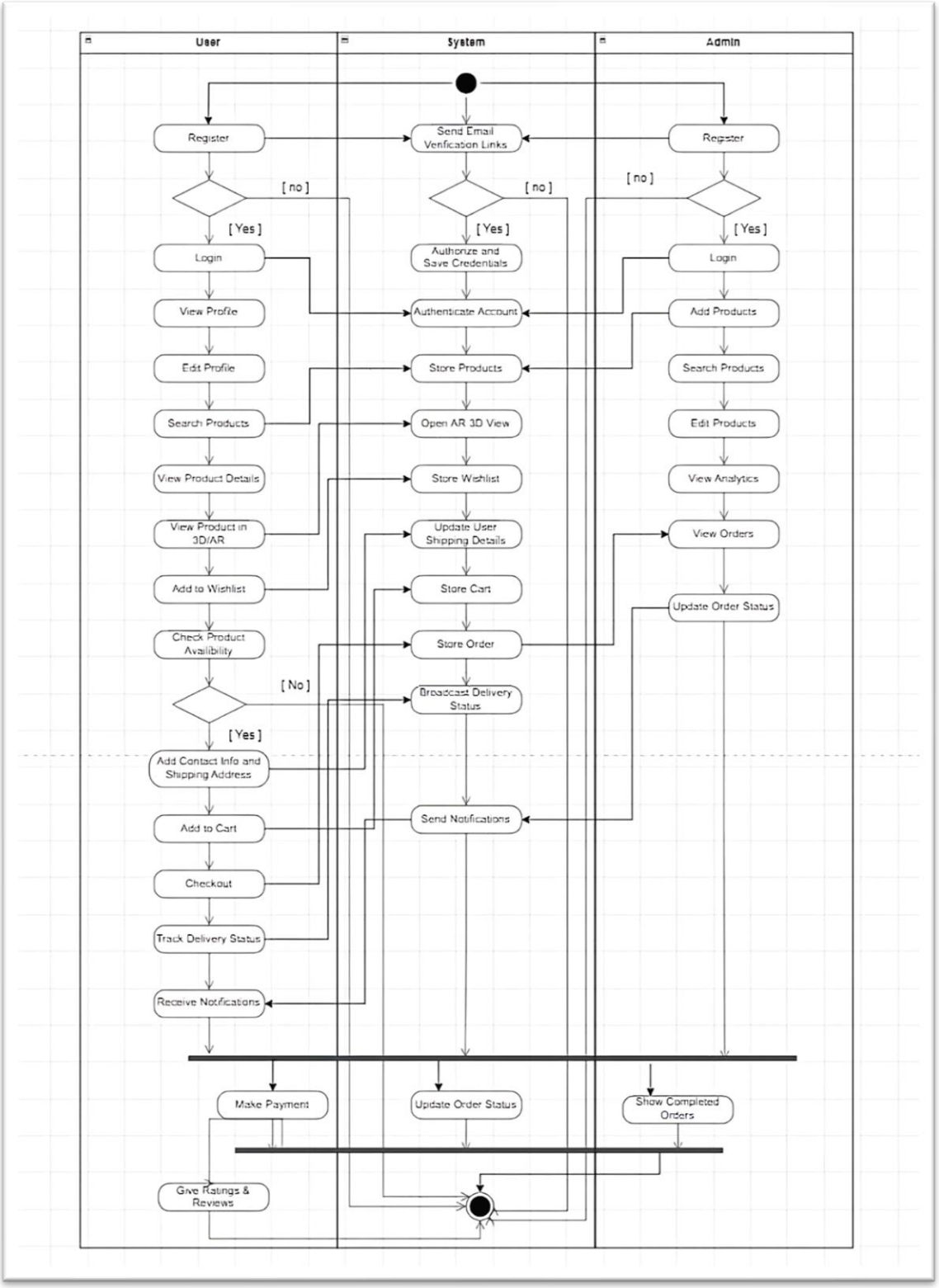
Decisions and Branching:
In Activity diagram, a diamond symbolizes a decision point with multiple paths. To depict activities that hinge on decisions, place a diamond between them. Each outgoing path should be labelled with a condition or guard expression.

Synchronisation:
A fork node in an activity diagram serves to divide a single incoming flow into multiple concurrent flows. It is symbolized as a straight, slightly thicker line, visually indicating the branching of parallel activities.

Swimlanes:
Swim lanes in activity diagrams group related activities into distinct columns, providing a visual organization that clarifies responsibilities or divisions within a process or system. Each lane typically represents a specific entity, role, or department involved in the activities.

Final State or End Point:
In activity diagrams, an arrow directed towards a filled circle enclosed within another circle symbolizes the final action state. This marks the conclusion of an activity or within the diagram's flow, providing clear visual cues for the endpoint.

Activity Diagram



NoSQL Schema Design

In a traditional SQL database, ERD diagrams are used to represent the structure of the data with entities (tables) and their relationships. However, in NoSQL databases, which include document-oriented databases like Firebase Firestore, graph databases, and key-value stores, the data structure is quite different. Therefore, instead of drawing an ERD (which is kind of wrong for Firebase projects because it is non-relational), we can represent the database using boxes.

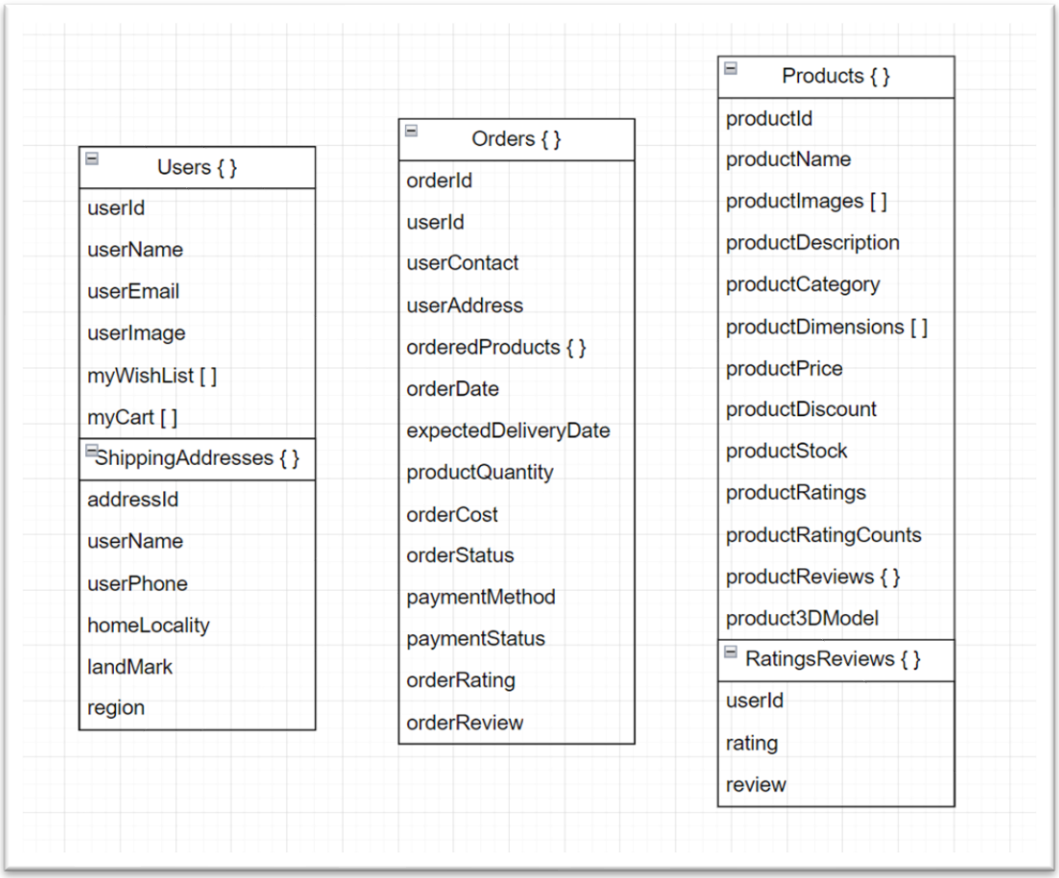
Entities as Collections:

In NoSQL databases, entities are typically organized as collections in document-based databases. These collections serve as containers for related data, similar to how tables represent entities in SQL databases. This flexible schema approach allows for the storage of diverse data structures within each collection, offering adaptability in data modeling.

Attributes as Fields:

In NoSQL databases, entities use fields to store data instead of traditional columns. These fields can accommodate diverse data types, including nested objects and arrays. This flexibility allows for the efficient storage of complex and varied data structures, making NoSQL databases suitable for handling diverse data models and unstructured information.

NoSQL Schema Design



Class Diagram

A class diagram is a fundamental concept in object-oriented modeling and design, often used in software engineering to represent the structure and relationships of classes within a system. Given your background in Android development using Kotlin and Google Firebase, class diagrams can be a useful tool for visualizing the architecture of your Android applications.

Classes:
In a class diagram, classes are represented as rectangles with three compartments. The top compartment typically contains the class name, the middle compartment lists the class attributes (variables), and the bottom compartment includes the class methods (functions).

Attributes:
Class attributes are the properties or variables that belong to a class. They're listed in the middle compartment of the class rectangle.

Methods:
Class methods represent the functions or operations that a class can perform. These are listed in the bottom compartment of the class rectangle.

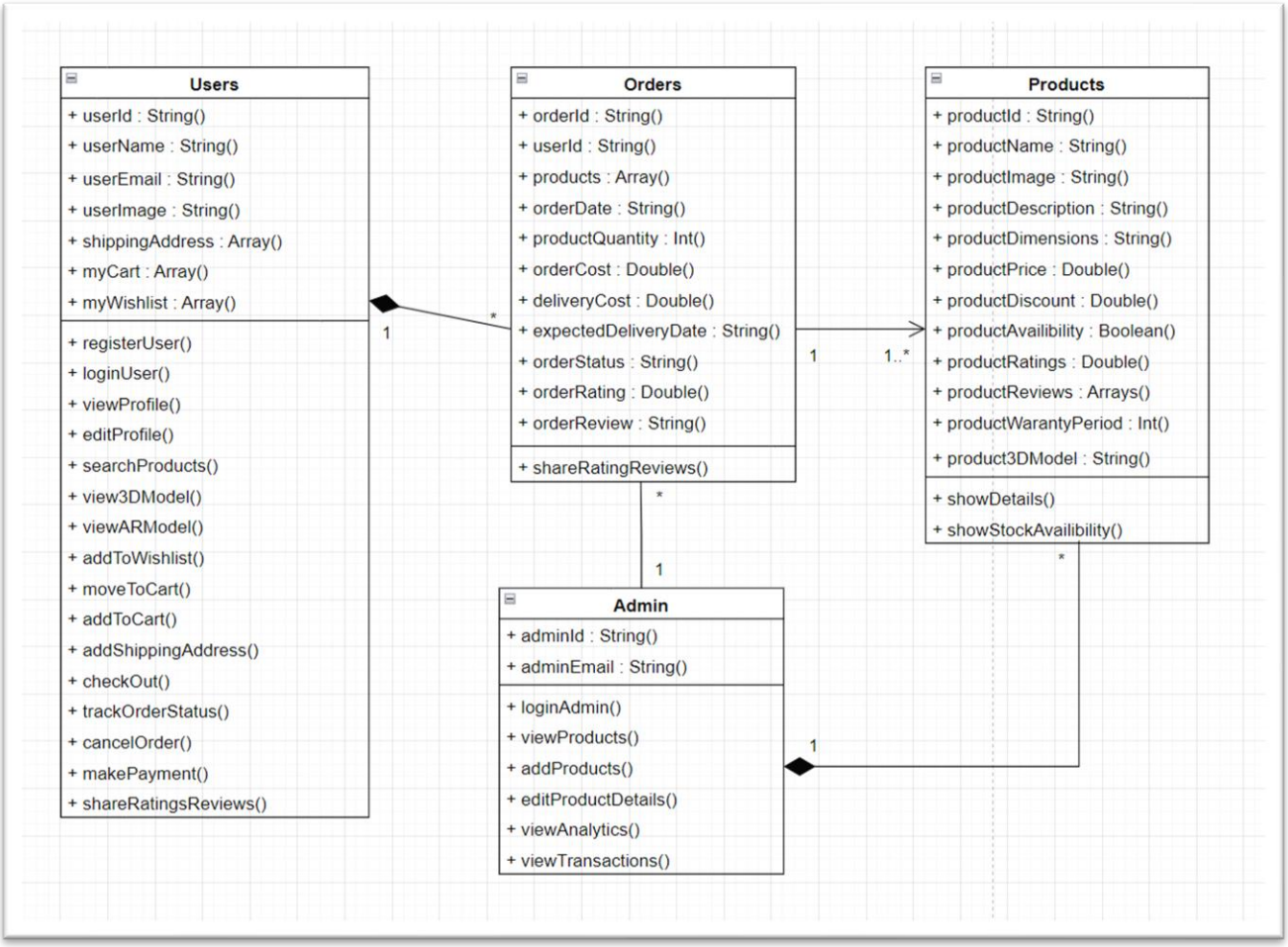
Associations:
In object-oriented programming and UML (Unified Modeling Language), there are several types of associations that represent different kinds of relationships between classes. Here are some common types of associations:

Association:
This is a basic relationship between classes, indicating that one class is associated with another in some way. It's a generic term for any connection between classes.

Aggregation:
Aggregation represents a "whole-part" relationship, where one class (the whole) contains or is composed of other classes (the parts). It's typically depicted as a diamond shape.

Composition:
Composition is a stronger form of aggregation, indicating that the parts are essential to the whole and can't exist independently. It's also depicted with a diamond shape.

Class Diagram



Sequence Diagram

A sequence diagram is a visual representation in software engineering that depicts the interactions between different components or objects within a system over a specific period. It illustrates the flow of messages or actions between these entities, helping developers understand the order and timing of these interactions. Sequence diagrams are commonly used to model and document the behaviour of software systems, making them a valuable tool in the design and analysis of complex systems.

In short, A sequence diagram is a UML (Unified Modeling Language) diagram that provides a detailed view of how various components or objects in a software system interact with each other.

Participants:

In a sequence diagram, participants are represented by lifelines, typically depicted as vertical lines. These lifelines represent objects, classes, or components involved in the interaction. Messages flow between these lifelines to represent interactions.

Messages:

Messages are depicted as arrows that flow between lifelines. They represent communication or interactions between objects. Messages can be synchronous, asynchronous, or create/destroy messages, depending on the type of interaction.

Activation Bar:

An activation bar is a horizontal bar along a lifeline that shows the period during which an object is active and processing a message. It helps visualize the timing of interactions.

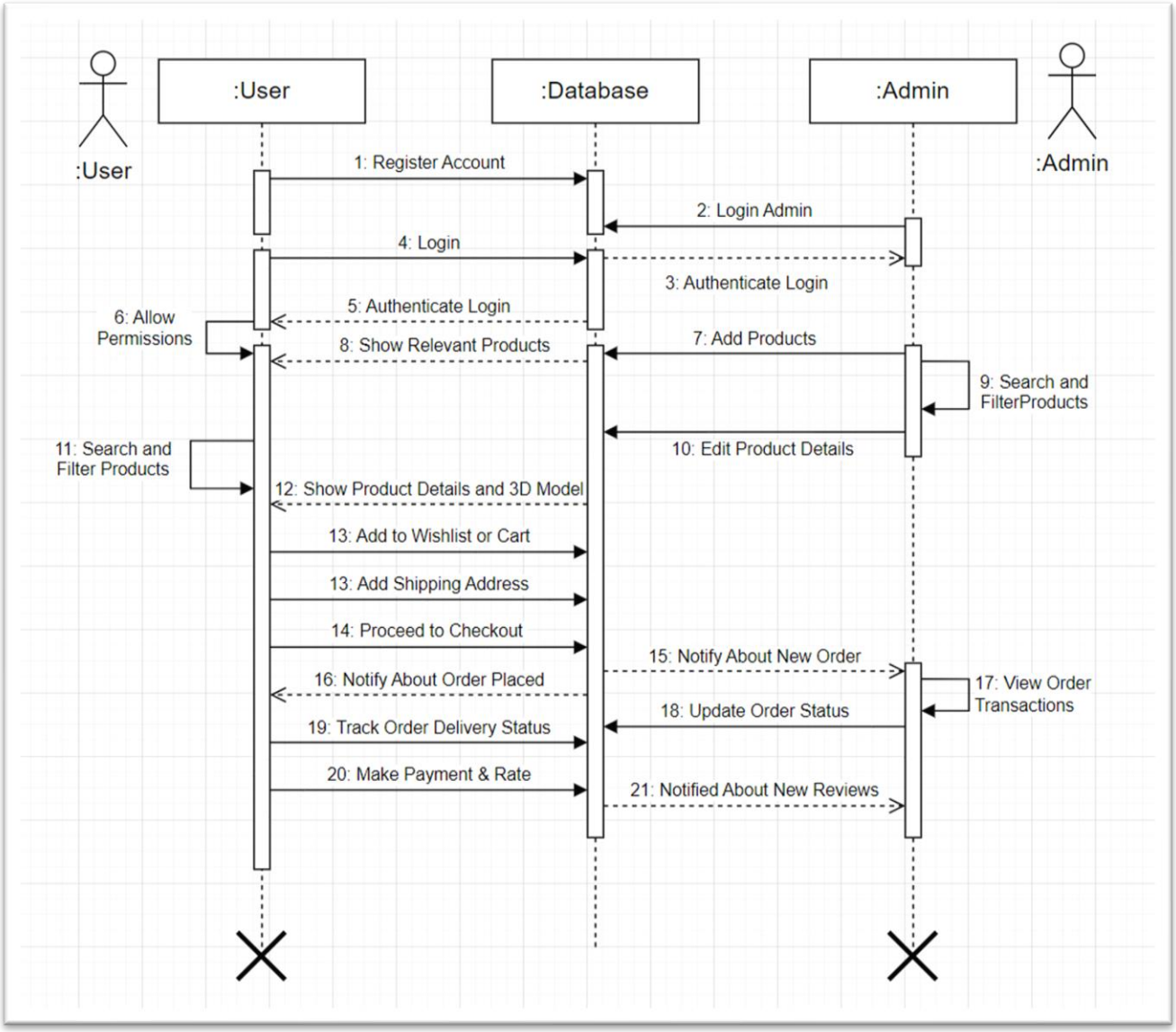
Focus on Time:

Sequence diagrams are especially useful for emphasizing the chronological order of interactions. Developers can see how messages are exchanged over time, aiding in understanding system behaviour and potential bottlenecks.

Lifelines:

Lifelines visually represent the lifespan or existence of the associated object or component within the sequence diagram. The top of the lifeline represents the creation of the object, and the bottom represents its destruction, if applicable.

Sequence Diagram



Test Cases

TestCaseID	Module	Form	Test Conditions	Steps	Input Test Data	Expected Output	Actual Output	Status
TC1	Registration	Registration Form	Provide valid Details by following all validations	1. Click on Sign-up Button 2. Fill Sign-up Details	1. Name 2. Email 3. Password 4. Re-Password	1. Registered Successfully 2. Verify Email Address	1. Registered Successfully 2. Verify Email Address	Pass
TC2	Login	Sign-in Form	Enter Valid Credentials	1. Click on Sign-in Button	1. Enter Email 2. Enter Password	Login Successful	Login Successful	Pass
TC3	Login	Sign-in Form	Enter Invalid Credentials	1. Enter Invalid Credentials	1. Enter Invalid Credentials	Invalid Credentials	Invalid Credentials	Pass
TC4	Google Sign-in	Sign-in Form	Select Valid Google Email ID	1. Select Google Account	1. Select Google Account	Login Successful	Login Successful	Pass
TC5	Forgot Password	Forgot Password Form	Enter Registered Email	1. Enter Registered Email	1. Enter Registered Email	Password Reset	Password Reset	Pass
TC6	Forgot Password	Forgot Password Form	Enter Unregistered Email	1. Enter Unregistered Email	1. Enter Unregistered Email	Email is not registered	Email is not registered	Pass
TC7	Login	Sign-in Form	User should be already Signed in	1. Sign in and Restart the App	Not Applicable	Directly Signs you in and home screen appears, no need for re-login	Directly Signs you in and home screen appears, no need for re-login	Pass
TC8	User	Home Page	Allow Permissions for notifications and read Images and Files from Local Storage	1. Allow Permissions	Not Applicable	Permissions Granted	Permissions Granted	Pass
TC9	User	Home Page	Navigate Products and click on add to wishlist button	1. Navigate to Home Page 2. Click on Add to WishList	Not Applicable	If Product not in wishlist then add it and vice versa	If Product not in wishlist then add it and vice versa	Pass

TC10	All	Offline Page	No Internet Connection / Device connected to WIFI but no internet	1. Turn off wifi/data connection	Not Applicable	Offline Page Appears.	Offline Page Appears	Pass
TC11	User	Profile Page	Click on Sign out Button	1. Go to Profile Page 2. Click on Sign out Button	Not Applicable	Return to Sign in Page	Return to Sign in Page	Pass
TC12	User	Profile Page	Open Profile Page & Internet Connection	1. Go to Profile Page	Not Applicable	Displays Details of the user	Displays Details of the user	Pass
TC13	User	Profile Page	Add Shipping Address	1. Open Profile Page 2. Click on add Shipping Address 3. Fill Details and Submit	1. Name 2. Contact Number 3.Local Address 4. Land Mark 5. Location	Shipping Address added successfully	Shipping Address added successfully	Pass
TC14	User	Profile Page	Edit user name and user image	1. Open Profile Page 2. Click on Edit button 3. Enter name and select image file to change username and userimage respectively	1. Username 2. Userimage file	Username and Userimage updated successfully	Username and Userimage updated successfully	Pass
TC15	Admin	Add Product Page	Add Product Details and select required Files	1. Open Add Product Page 2. Add details and upload required files	1. Product name 2. Price 3. Discount 4. Stock 5. 3Dmodel 6.Description 7. Dimensions,etc.	Product Added Successfully	Product Added Successfully	Pass
TC16	Admin	Search Products page	Enter Product	1. Search a product	1. Product Name	Product Details	Product Details	Pass

			Details to change	2. Select a product 3. Click on edit button 4. Enter Details 5. Update product	2. Product Description 3. Product price 4. Product Discount 5. Product Stock	Updated Succesfully	Updated Successfully	
TC17	User	Product Details Page	Open Product Details Page	1. Select a product to open its detailed view	Not Applicable	All Product Details along with various options like add to cart, buynow,etc will be shown	All Product Details along with various options like add to cart, buynow,etc will be shown	Pass
TC18	User	Product Details Page	Open Product Details Page	1. Select a product to open its detailed view 2. Click on Add to Cart Button	Not Applicable	Adds the Current Product to Cart	Adds the Current Product to Cart	Pass
TC19	User	Product Search Page	Open Products Search Page	1. Open garage Search Page 2. Enter Product name to search 3. Apply filters to search and categorize products	1. Search Text or the name of the product 2. Select the Filter that you want to apply	Displays various products depending upon the search text and filter applied	Displays various products depending upon the search text and filter applied	Pass
TC20	User	Product Details Page	Select a Product to view its details and click AR and 3D View button	1. Select a product to open 2. Click on AR and 3D View Button	Not Applicable	Product's 3D model and AR view is shown	Product's 3D model and AR view is shown	Pass
TC21	User	Wish list Page	Open wish list page and some products are there in wish list and	1.Open wish list page 2. Click on move all products to wishlist	Not Applicable	All Products moved to cart	All products moved to cart	Pass

			click on move all products to cart					
TC22	User	My Cart Page	Open My Cart Page and Products are there in my cart and then Click on Proceed to Checkout	1.Open My Cart Page 2. Click on Proceed to Checkout	Not Applicable	Products in Cart are moved to Checkout Page	Products in Cart are moved to Checkout Page	Pass
TC23	User	Wish List Page	Open wish list with no products added in it and click on move all products to Cart	1. Open Wish List Page 2. Click on Move all products to Cart	Not Applicable	Wish list is Empty	Wish list is Empty	Pass
TC24	User	My Cart Page	Open My Cart Page with no products in it and click on Proceed to checkout	1. Open My Cart Page 2. Click on Proceed to Checkout Page	Not Applicable	My Cart is Empty cannot proceed to checkout	My Cart is Empty cannot proceed to checkout	Pass
TC25	User	Profile Page	Open profile page and open my orders	1. Open Profile Page 2. Click on My Orders	Not Applicable	All Orders will be shown in a compact view with categorization like order progress, order delivered and order cancelled	All Orders will be shown in a compact view with categorization like order progress, order delivered and order cancelled	Pass
TC26	User	Orders Page	Open Orders and select order to get a detailed view of it.	1. Open Profile Page 2. Go to My Orders 3. Select a order to get a detailed view	Not Applicable	A detailed view of the order is shown with all required details	A detailed view of the order is shown with all required details	Pass

TC27	User	Orders Page	Open My Orders Page and open a order which is delivered	1. Open Profile Page 2. Go to My Orders 3. Select a orders which is delivered 4. Add a Rating and Review to the order	1. Rating 2.Review	Rating and Review gets submitted and notification comes that it is submitted	Rating and Review gets submitted notifications comes that it is submitted	Pass
TC28	User	My Cart Page	Open My Cart Page and there are products in the cart. Click on increment the quantity of the product	1. Open My Cart Page 2. Increment or Decrement the quantity of products but click on add or minus button.	1. Select a Product to Update its Quantity	Product Quantity is incremented or decremented and accordingly the overall prices and discounts are also changed in real-time	Product Quantity is incremented or decremented and accordingly the overall prices and discounts are also changed in real-tiime	Pass
TC29	User	Wish List Page	Open wish list page and there are items in it. Click on remove items to remove them from the wish list	1. Open wish list page 2. Click on remove from wish list button on a particular product	Not Applicable	Selected Product will be removed from the wish list	Selected Product will be removed from the wish list	Pass
TC30	User	Profile Page	Click on Reset Password to get a reset password link in to your email	1. Open Profile page 2. Enter Registered Email and Click on Reset Password	1. New Password 2. Re-type New Password	Password Changed Successfully	Password Changed Succesfully	Pass
TC31	User	My Cart Page	Open My Cart Page and there are items in my cart. Click on remove	1. Open my cart page 2. Click on remove all items from my cart	Not Applicable	All the items from the cart are removed successfully	All the items from the cart are removed successfully	Pass

			all items from my cart					
TC32	Admin	Order Transactions Page	Open Transactions Page as admin and select a product to update its status if it is not cancelled.	1. Open Transactions Page 2. Check for ongoing orders transactions 3. Click on it to update its status	Not Applicable	The selected order status will be updated and the user of that particular order will be notified	The selected order status will be updated and the user of the particular order will be notified	Pass
TC33	Admin	Admin Insights Page	Open admin Insights page	1. Open admin insights page	Not Applicable	Relevant Visualized graphs will be displayed	Relevant Visualized graphs will be displayed	Pass
TC34	User	Orders Page	Open Orders page and selected a order that is not delivered	1. Open Orders Page 2. Select a order which is not delivered	Not Applicable	The Rating Review buttons will be disabled until the order's status changes to order delivered	The Rating Review buttons will be disabled until the order's status changes to order delivered	Pass

Component Diagram

A component diagram is a type of UML (Unified Modeling Language) diagram used in software engineering to visualize the high-level structure and organization of a software system or application. While it may not be directly related to your Android development and Firebase expertise, understanding component diagrams can still be valuable in the context of software design and architecture.

A component diagram is primarily focused on depicting the various components or modular units that make up a software system and how they interact with each other. These components can represent classes, packages, libraries, or even physical modules, depending on the level of abstraction you want to depict.

Components:

Components are the fundamental building blocks of a system. They encapsulate a set of related functionalities and can be represented as rectangles in the diagram. Each component typically corresponds to a software module or a class.

Composite Components:

In larger systems, it's common to group related components into higher-level composite components. These composite components can be represented as larger rectangles containing other components, showing a higher level of abstraction.

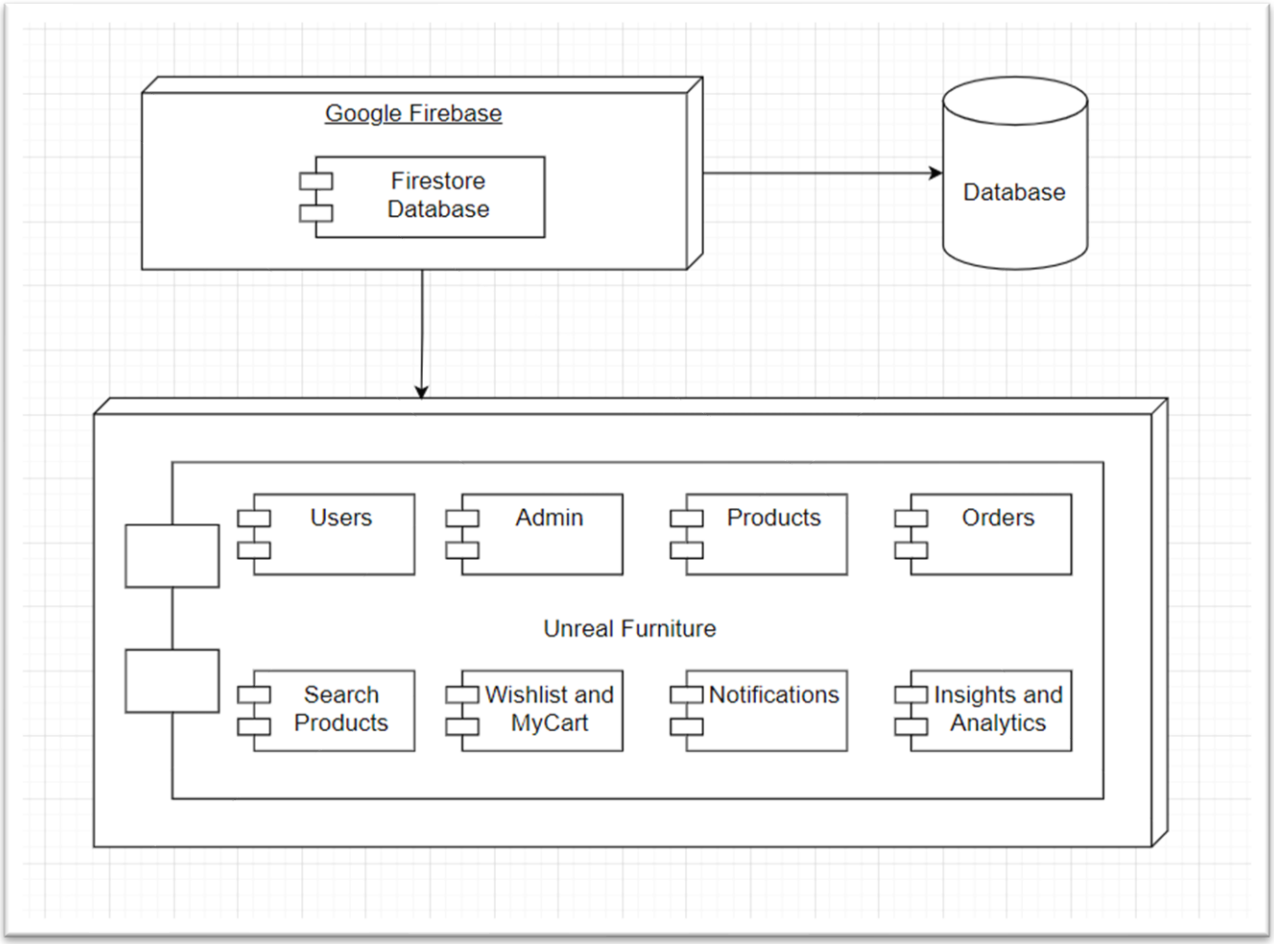
Database Component:

Represent the database as a separate component within your component diagram. This component should have a name that clearly identifies it as the database, such as "Database" or the name of the specific database system you are using, like "MySQL," "Firebase Firestore", etc.

Connectors:

Connectors are used to depict the relationships between components and interfaces. They define how data or control flows between components.

Component Diagram



Deployment Diagram

A deployment diagram is a type of diagram in the Unified Modeling Language (UML) that is used to visualize and describe the architecture of a system as it is deployed in a real-world environment. Deployment diagrams can be valuable for understanding how your mobile app interacts with backend services, servers, and other components when it's deployed in a production environment.

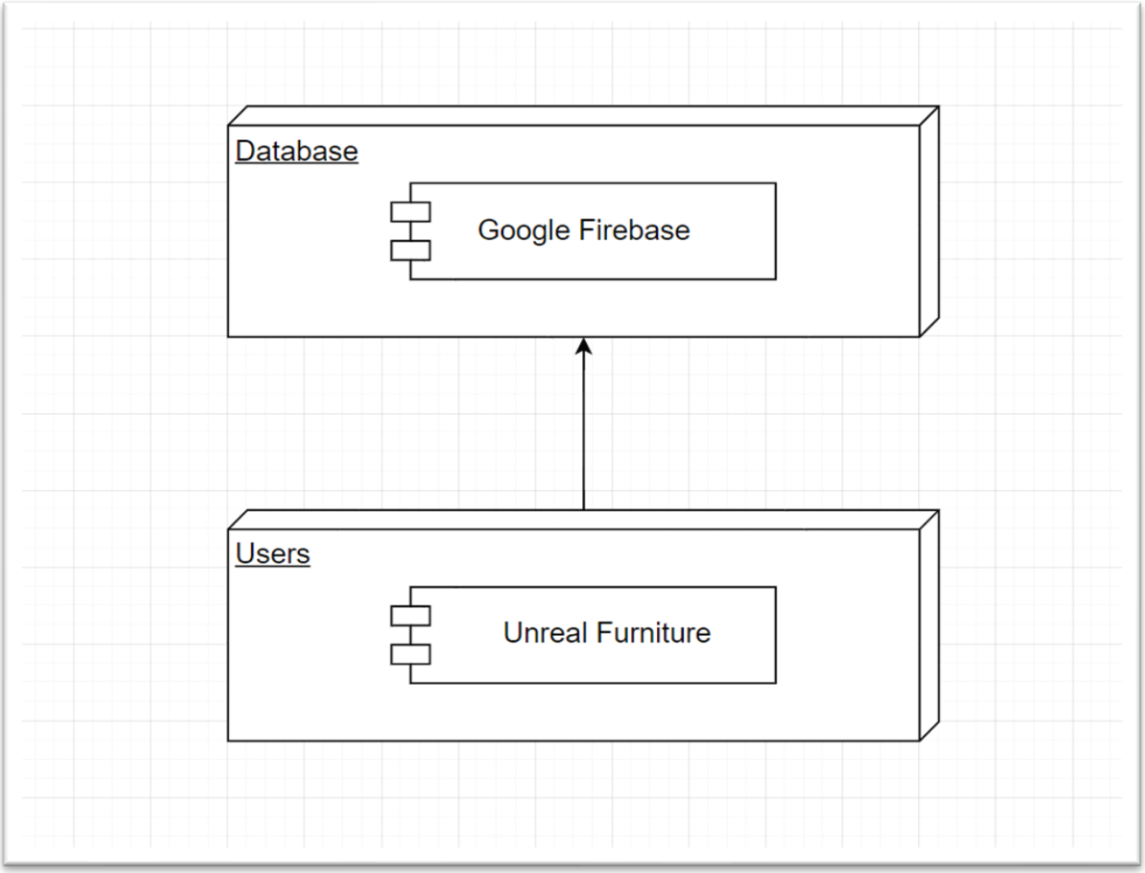
Nodes:

Nodes represent physical or virtual computational resources in your deployment. In your case, these could be servers, cloud services, or even mobile devices. For example, you might have a node for your Android devices and another for the Firebase server.

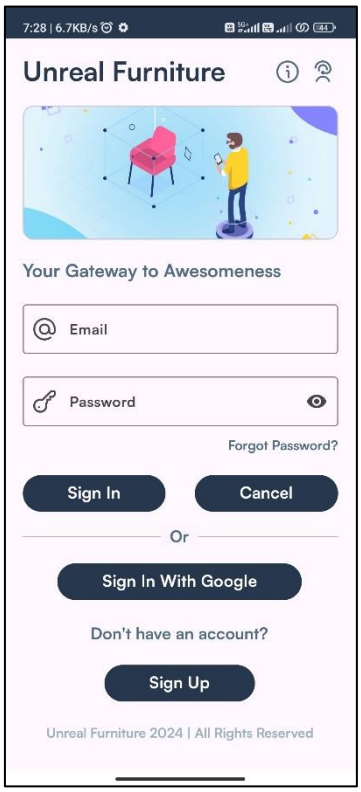
Dashed Lines:

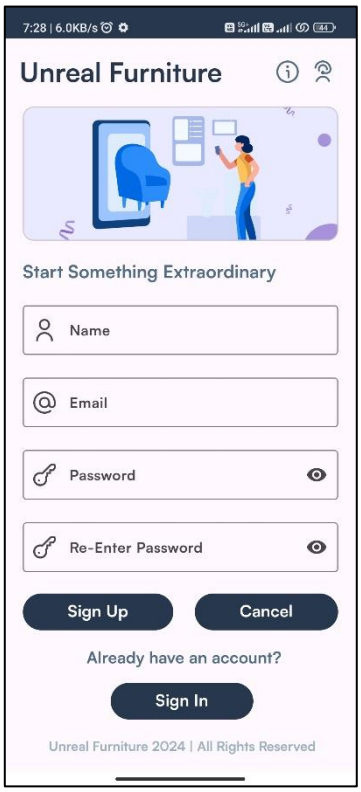
The dependency on node on components is depicted using dashed lines. This means that a node uses the services of the components that are executing on another node.

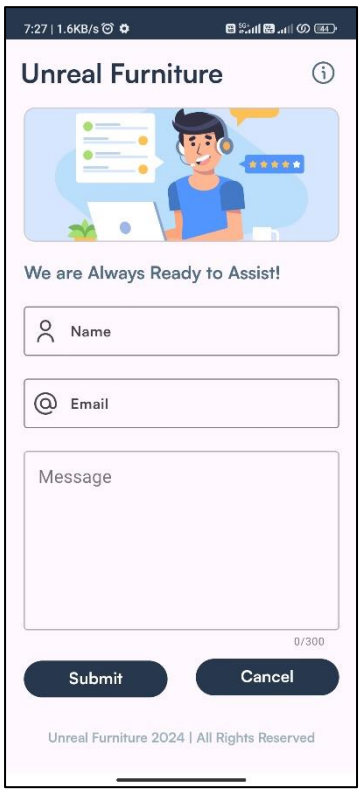
Deployment Diagram

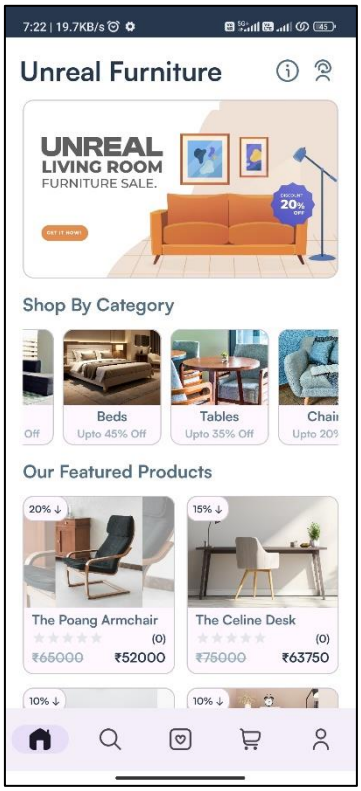


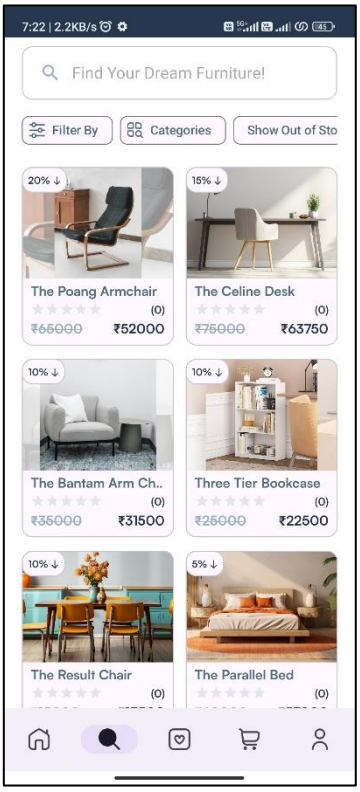
Images of Application

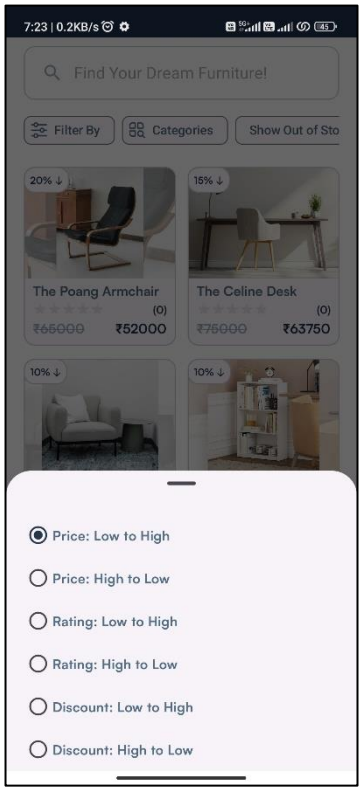








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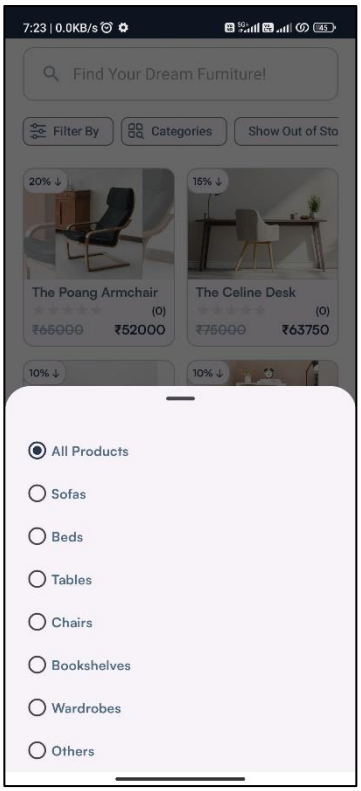
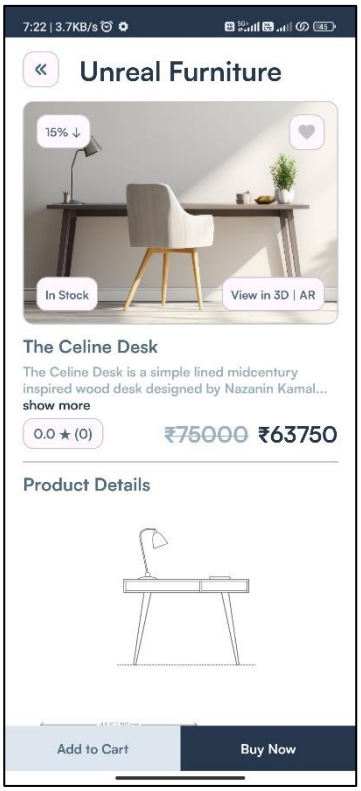
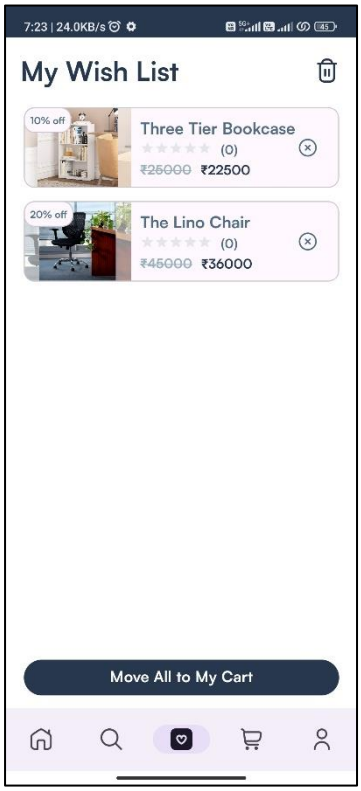
Find Your Dream Furniture!
Filter By
Categories
Show Out of Sto
20% ↓
15% ↓
The Poang Armchair
(0)
₹65000 ₹52000
The Celine Desk
(0)
₹75000 ₹63750
10% ↓
10% ↓
Price: Low to High
Price: High to Low
Rating: Low to High
Rating: High to Low
Discount: Low to High
Discount: High to Low


7:23 | 0.0KB/s
Find Your Dream Furniture!
Filter By
Categories
Show Out of Sto
20% ↓
15% ↓
The Poang Armchair
(0)
₹65000 ₹52000
The Celine Desk
(0)
₹75000 ₹63750
10% ↓
10% ↓
All Products
Sofas
Beds
Tables
Chairs
Bookshelves
Wardrobes
Others


7:22 | 3.7KB/s
Unreal Furniture
15% ↓
In Stock
View in 3D | AR
The Celine Desk
The Celine Desk is a simple lined midcentury inspired wood desk designed by Nazanin Kamal...
show more
0.0 ★ (0)
₹75000 ₹63750
Product Details
Add to Cart
Buy Now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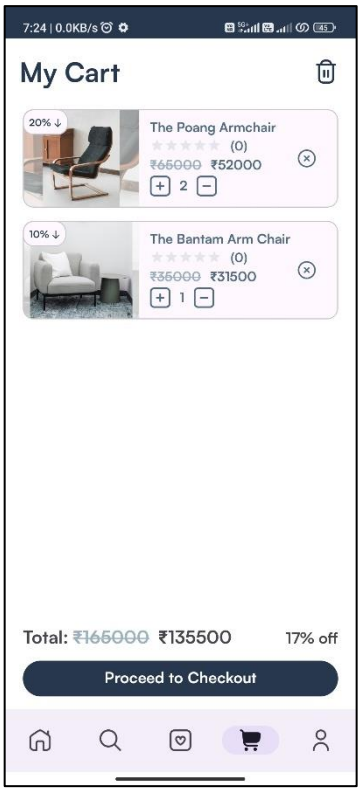
My Wish List
10% off
Three Tier Bookcase
(0)
₹25000 ₹22500
20% off
The Lino Chair
(0)
₹45000 ₹36000
Move All to My Cart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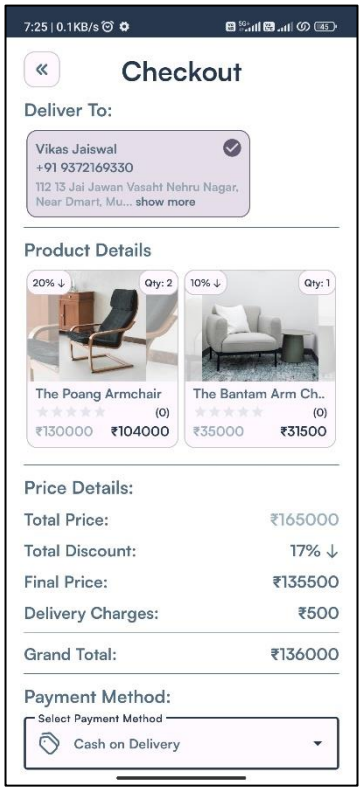
My Cart
20% ↓
The Poang Armchair
(0)
₹65000 ₹52000
2
10% ↓
The Bantam Arm Chair
(0)
₹35000 ₹31500
1
Total: ₹165000 ₹135500
17% off
Proceed to Checkout

7:25 | 5.4KB/s
My Cart
20% ↓
The Poang Armchair
(0)
₹65000 ₹52000
2
10% ↓
The Bantam Arm Chair
(0)
₹35000 ₹31500
1
Check Out
Are you sure you want to check out?
No
Yes
Total: ₹165000 ₹135500
17% off
Proceed to Checkout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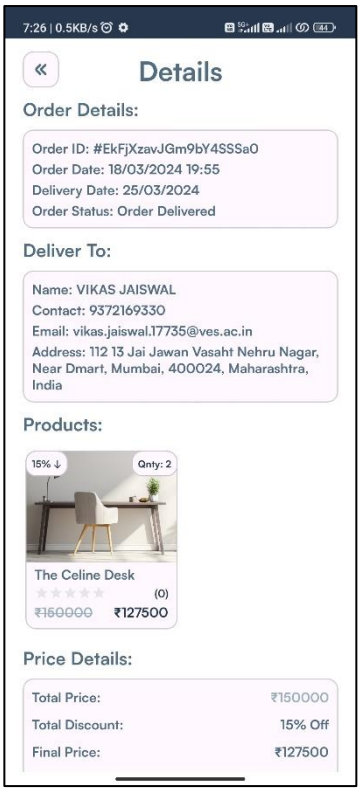
Checkout
Deliver To:
Vikas Jaiswal
+91 9372169330
112 13 Jai Jawan Vasaht Nehru Nagar, Near Dmart, Mu... show more
Product Details
20% ↓
Qty: 2
The Poang Armchair
(0)
₹130000 ₹104000
10% ↓
Qty: 1
The Bantam Arm Ch..
(0)
₹35000 ₹31500
Price Details:
Total Price: ₹165000
Total Discount: 17% ↓
Final Price: ₹135500
Delivery Charges: ₹500
Grand Total: ₹136000
Payment Method:
Select Payment Method
Cash on Delivery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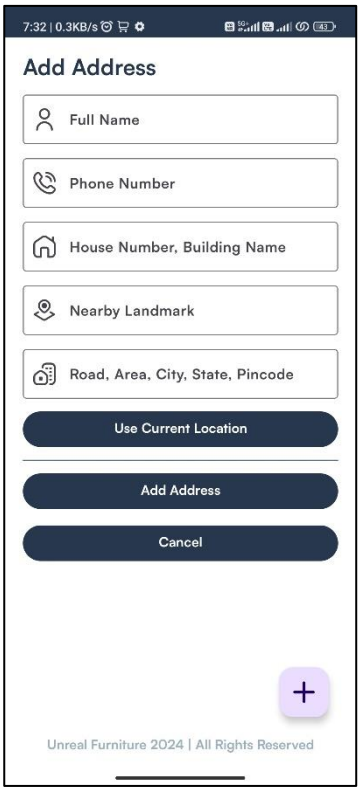
Details
Order Details:
Order ID: #EkFjXzavJGm9bY4SSSa0
Order Date: 18/03/2024 19:55
Delivery Date: 25/03/2024
Order Status: Order Delivered
Deliver To:
Name: VIKAS JAISWAL
Contact: 9372169330
Email: vikas.jaiswal.17735@ves.ac.in
Address: 112 13 Jai Jawan Vasaht Nehru Nagar, Near Dmart, Mumbai, 400024, Maharashtra, India
Products:
15% ↓
Qnty: 2
The Celine Desk
(0)
₹150000 ₹127500
Price Details:
Total Price: ₹150000
Total Discount: 15% Off
Final Price: ₹127500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Add Address
Full Name
Phone Number
House Number, Building Name
Nearby Landmark
Road, Area, City, State, Pincode
Use Current Location
Add Address
Cancel
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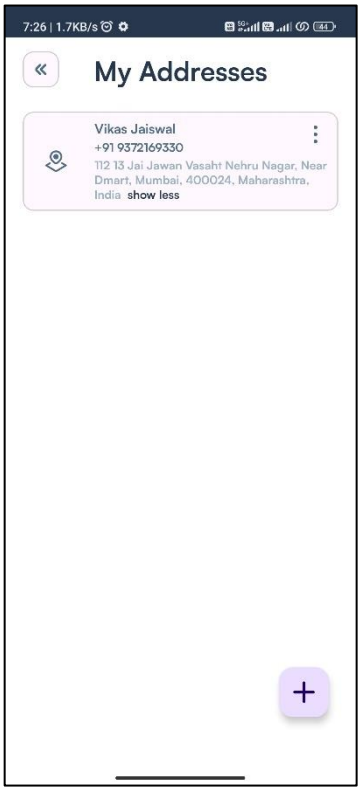
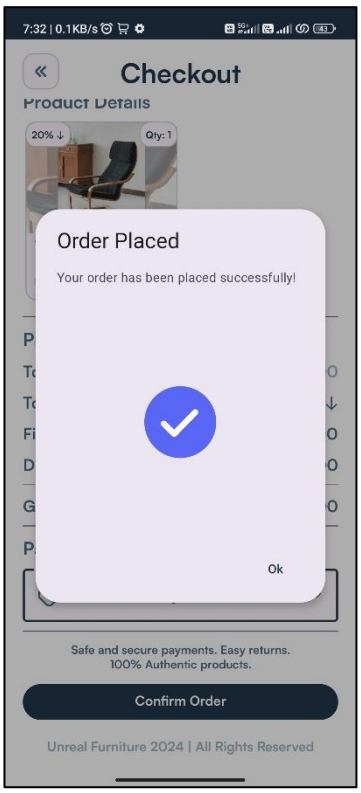
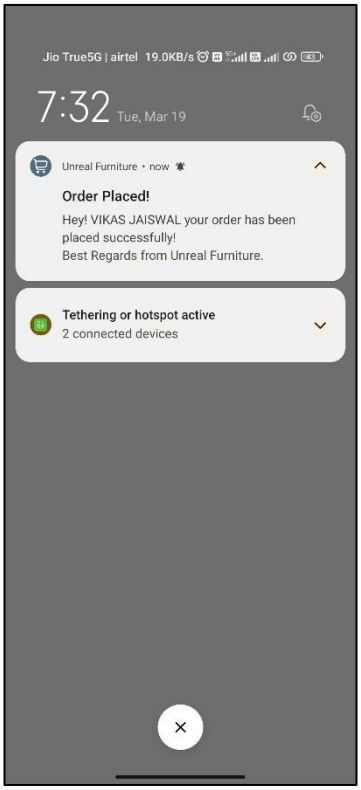
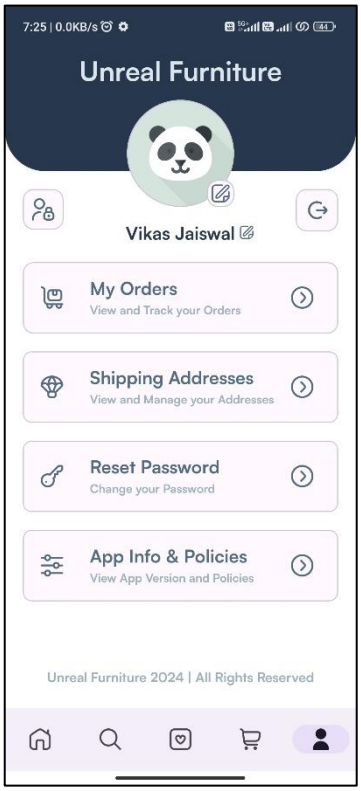
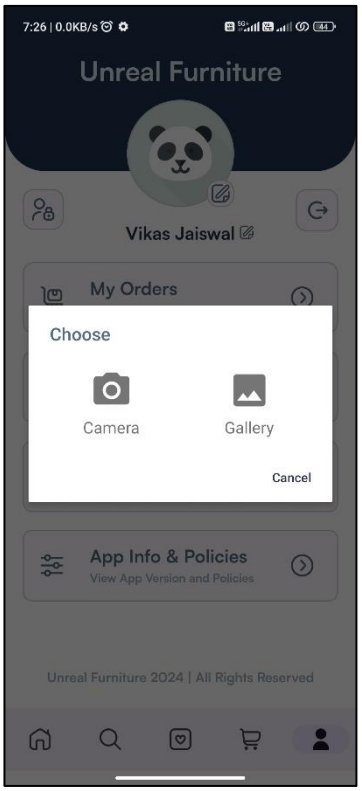

7:26 | 1.7KB/s
My Addresses
Vikas Jaiswal
+91 9372169330
112 13 Jai Jawan Vasaht Nehru Nagar, Near Dmart, Mumbai, 400024, Maharashtra, India show less


7:32 | 0.1KB/s
Checkout
Product Details
20% ↓
Qty: 1
Order Placed
Your order has been placed successfully!
Ok
Safe and secure payments. Easy returns.
100% Authentic products.
Confirm Order
Unreal Furniture 2024 | All Rights Reserved

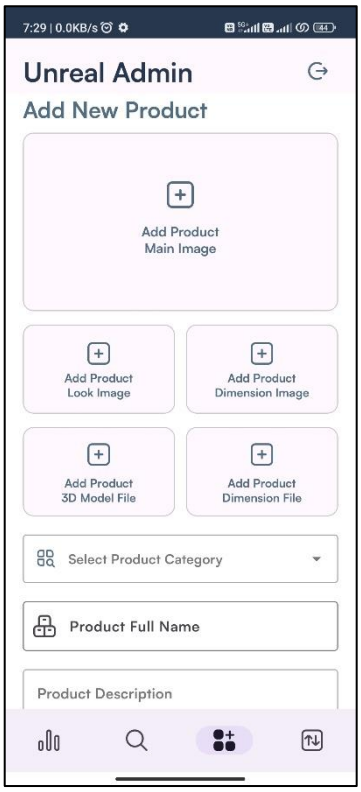

Jio True5G | airtel 19.0KB/s
7:32 Tue, Mar 19
Unreal Furniture • now
Order Placed!
Hey! VIKAS JAISWAL your order has been placed successfully!
Best Regards from Unreal Furniture.
Tethering or hotspot active
2 connected devices


7:25 | 0.0KB/s
Unreal Furniture
Vikas Jaiswal
My Orders
View and Track your Orders
Shipping Addresses
View and Manage your Addresses
Reset Password
Change your Password
App Info & Policies
View App Version and Policies
Unreal Furniture 2024 | All Rights Reserved

7:26 | 0.1KB/s
Unreal Furniture
Vikas Jaiswal
My Orders
New User Name
Name
Vikas Jaiswal
Cancel
Save
App Info & Policies
View App Version and Policies
Unreal Furniture 2024 | All Rights Reserved


7:26 | 0.0KB/s
Unreal Furniture
Vikas Jaiswal
My Orders
Choose
Camera
Gallery
Cancel
App Info & Policies
View App Version and Policies
Unreal Furniture 2024 | All Rights Reserved

8:11 | 171KB/s
Unreal Furniture
Unreal Furniture 2024 | All Rights Reserved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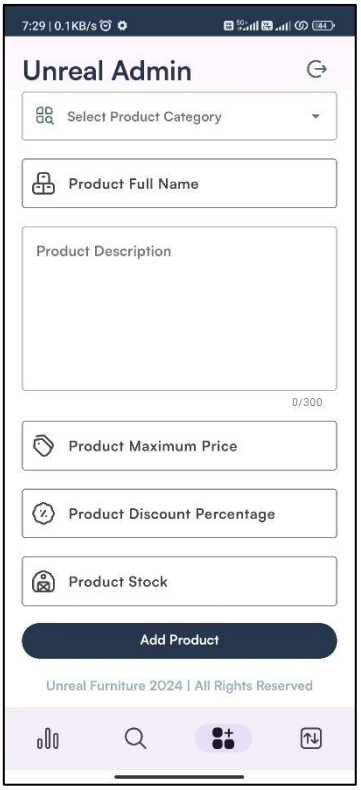
Unreal Admin
Add New Product
Add Product Main Image
Add Product Look Image
Add Product Dimension Image
Add Product 3D Model File
Add Product Dimension File
Select Product Category
Product Full Name
Product Description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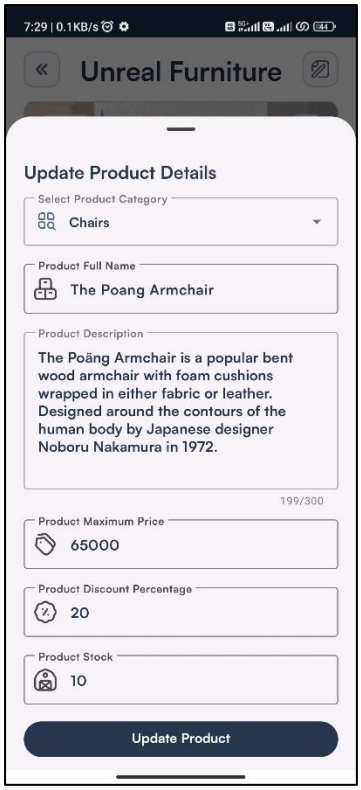
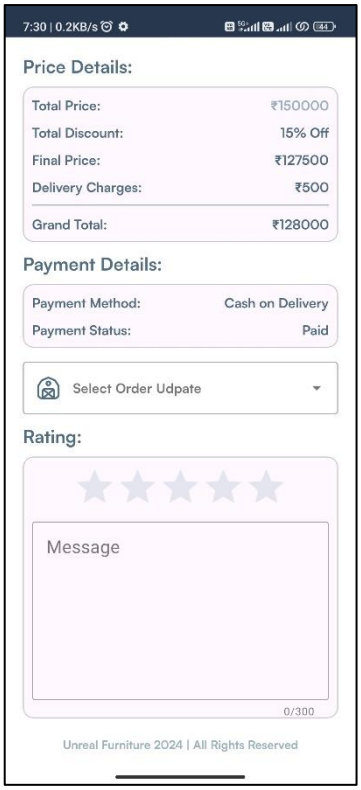
Unreal Admin
Select Product Category
Product Full Name
Product Description
0/300
Product Maximum Price
Product Discount Percentage
Product Stock
Add Product
Unreal Furniture 2024 | All Rights Reserved


7:29 | 0.1KB/s
Unreal Furniture
Update Product Details
Select Product Category
Chairs
Product Full Name
The Poang Armchair
Product Description
The Poäng Armchair is a popular bent wood armchair with foam cushions wrapped in either fabric or leather. Designed around the contours of the human body by Japanese designer Noboru Nakamura in 1972.
199/300
Product Maximum Price
65000
Product Discount Percentage
20
Product Stock
10
Update Product


7:30 | 0.2KB/s
Price Details:
Total Price: ₹150000
Total Discount: 15% Off
Final Price: ₹127500
Delivery Charges: ₹500
Grand Total: ₹128000
Payment Details:
Payment Method: Cash on Delivery
Payment Status: Paid
Select Order Udpate
Rating:
Message
0/300
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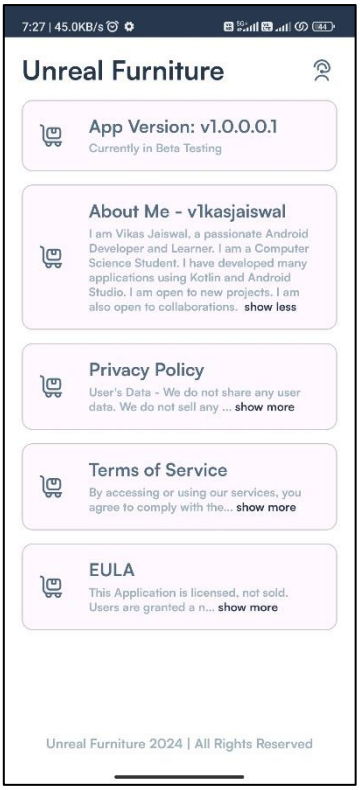
7:27 | 45.0KB/s
Unreal Furniture
App Version: v1.0.0.0.1
Currently in Beta Testing
About Me - v1kasjaiswal
I am Vikas Jaiswal, a passionate Android Developer and Learner. I am a Computer Science Student. I have developed many applications using Kotlin and Android Studio. I am open to new projects. I am also open to collaborations. show less
Privacy Policy
User's Data - We do not share any user data. We do not sell any ... show more
Terms of Service
By accessing or using our services, you agree to comply with the... show more
EULA
This Application is licensed, not sold. Users are granted a n... show more
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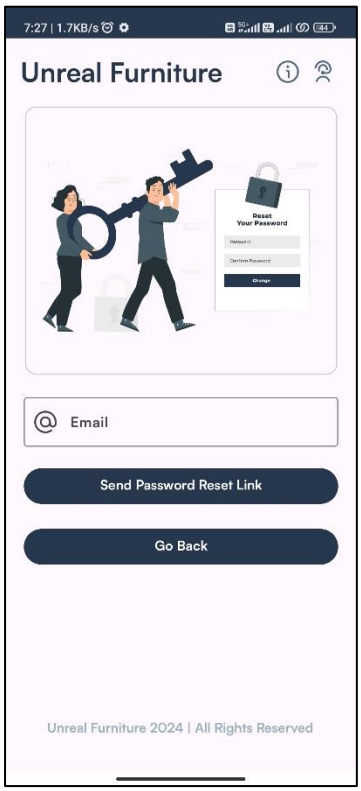
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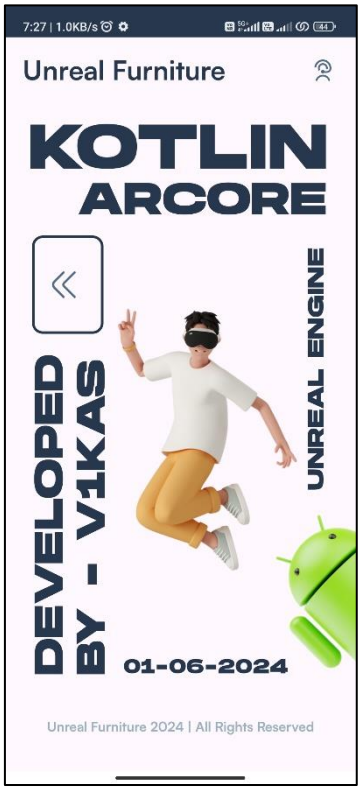
Unreal Furniture
Email
Send Password Reset Link
Go Back
Unreal Furniture 2024 | All Rights Reserved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Unreal Furniture
KOTLIN
ARCORE
DEVELOPED BY - V1KAS
UNREAL ENGINE
01-06-2024
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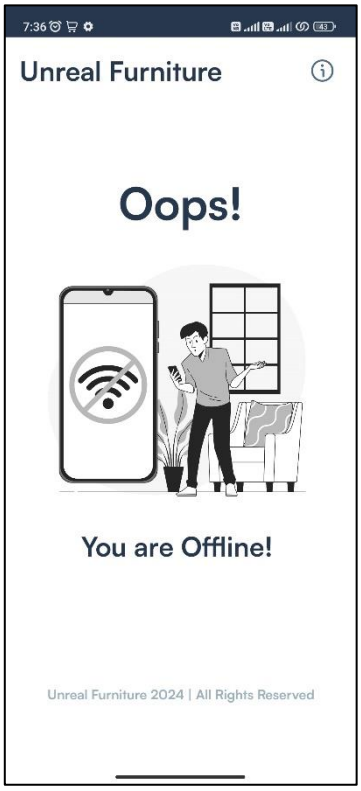
7:36
Unreal Furniture
Oops!
You are Offline!
Unreal Furniture 2024 | All Rights Reserved

Future Enhancements

Advanced Search Filtering:

Advanced search filters for finding the best furniture products, using Firebase's Firestore database to query and filter results based on criteria like fabric, material, colors, location, etc.

Enhanced Recommendation System:

Users will be recommended with various products depending upon their last purchases, search history, preferences and wishlists.

Offline Mode:

Allow users to access basic functionalities even when they have limited or no internet connectivity, ensuring usability in areas with poor network coverage.

Improved 3D and AR View:

Show users an improved 3D model and AR view using AR Core and Unreal Engine so that even textures of the furniture are also rendered enhancing the user experience.

Product Management Tools:

Enhance the Admin dashboard with various tools for better business management, such as inventory tracking and employee scheduling.

References

Source: https://medium.com/firebase-developers/android-firebase-authentication-with-google-1c2f6ca3a738

Source: https://github.com/Dhaval2404/ImagePicker

Source: https://aayushpuranik.medium.com/search-view-android-2638c493693b

Source: https://www.geeksforgeeks.org/create-and-add-data-to-firebase-firestore-in-android/

Source: https://www.geeksforgeeks.org/android-recyclerview-in-kotlin/

Source: https://youtu.be/L_6poZGNXOo?si=s1FfKC9PHvsNeze4

Source: https://androidgeek.co/notifications-in-kotlin-beginner-in-depth-guide-13fa7d0748fc